2025 Summer Qual

II.

(a) Using the hint, $A = BC$

$$A^2 = A \Leftrightarrow BCBC = BC$$

$$\Rightarrow \exists D \text{ s.t. } CD = I \quad \because \text{ full row rank}$$

$$\Rightarrow BCBCD = BCD$$
$$BCB = B$$

$$\Rightarrow \exists L \text{ s.t. } LB = I \quad \because \text{ full column rank}$$

$$\Rightarrow LBCB = LB$$
$$CB = I$$

$$\begin{aligned} tr(A) = tr(A^2) &= tr(BCBC) \\ &= tr(CBCB) \\ &= tr(I_r) \\ &= r =: \text{rank}(A) \end{aligned}$$

$\blacksquare$

(b) Let G_1 and G_2 s.t. $AG_1A^TA = A$,
$AG_2A^TA = A$

$$0 = (AG_1A^TA - AG_2A^TA)(G_1^TA^T - G_2^TA^T)$$
$$= \underbrace{(AG_1A^T - AG_2A)}_{=B^T}(AG_1A^T - AG_2A^T)^T$$

$= B^T B$

$\Rightarrow B = 0$

$\Rightarrow AG_1A^T - AG_2A = 0 \rightsquigarrow AG_1A^T = AG_2A^T$

So, $A(A^TA)^-A^T$ is invariant to the choice of $(A^TA)^-$. $\square$

$\boxed{2}$.

$X_2^* = (I - H_{X_1})X_2, \quad H_{X_1} = X_1(X_1^TX_1)^{-1}X_1^T$

$Y = X_1\beta_1 + X_2^*\beta_2 + \varepsilon.$

$$(X^TX)^{-1} = \begin{pmatrix} X_1^TX_1 & \overbrace{X_1^TX_2^*}^{=0} \\ \underbrace{X_2^{*T}X_1}_{=0} & X_2^{*T}X_2^* \end{pmatrix}^{-1}$$

$$= \begin{pmatrix} (X_1^TX_1)^{-1} & 0 \\ 0 & (X_2^{*T}X_2^*)^{-1} \end{pmatrix}$$

$\widetilde{\beta_1} = (X_1^TX_1)^{-1}X_1^TY$

$\mathbb{E}[\widetilde{\beta_1}] = (X_1^TX_1)^{-1}X_1^T[X_1\beta_1 + X_2\beta_2]$

$= \beta_1 + (X_1^TX_1)^{-1}X_1^TX_2\beta_2$

① bias of $\widetilde{\beta_1} = (X_1^TX_1)^{-1}X_1^TX_2\beta_2$

$$\mathrm{var}(\hat{\beta}_1) = \sigma^2 (X_1^T X_1)^{-1}$$

$$\mathrm{var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$$

$$= \sigma^2 \begin{pmatrix} \boxed{X_1^T X_1} & X_1^T X_2 \\ X_2^T X_1 & X_2^T X_2 \end{pmatrix}^{-1}$$

$$= \sigma^2 \begin{pmatrix} (X_1^T X_1)^{-1} + (X_1^T X_1)^{-1} X_1^T X_2 \left(X_2^T (I - H_{X_1}) X_2\right)^{-1} X_2^T X_1 (X_1^T X_1)^{-1} & \star \\ \star & \star \end{pmatrix}$$

$$\mathrm{var}(\hat{\beta}_1) = \sigma^2 \left[(X_1^T X_1)^{-1} + (X_1^T X_1)^{-1} X_1^T X_2 \left(X_2^T (I - H_{X_1}) X_2\right)^{-1} X_2^T X_1 (X_1^T X_1)^{-1} \right]$$

$$\mathrm{var}(\hat{\beta}_1) - \mathrm{var}(\tilde{\beta}_1)$$

$$= \sigma^2 \underbrace{(X_1^T X_1)^{-1} X_1^T X_2}_{A} \underbrace{\left(X_2^T (I - H_{X_1}) X_2\right)^{-1}}_{B^{-1}} \underbrace{X_2^T X_1 (X_1^T X_1)^{-1}}_{A^T}$$

$I - H_{X_1}$ is symm & idempotent

$$\Rightarrow X_2^T (I - H_{X_1}) X_2 = X_2^T L L^T X_2 \quad \because \text{Cholesky decomposition}$$

$$= (L^T X_2)^T (L^T X_2) \quad \therefore \text{positive semidefinite}$$

② $\mathrm{var}(\hat{\beta}_1) - \mathrm{var}(\tilde{\beta}_1) \succeq 0$

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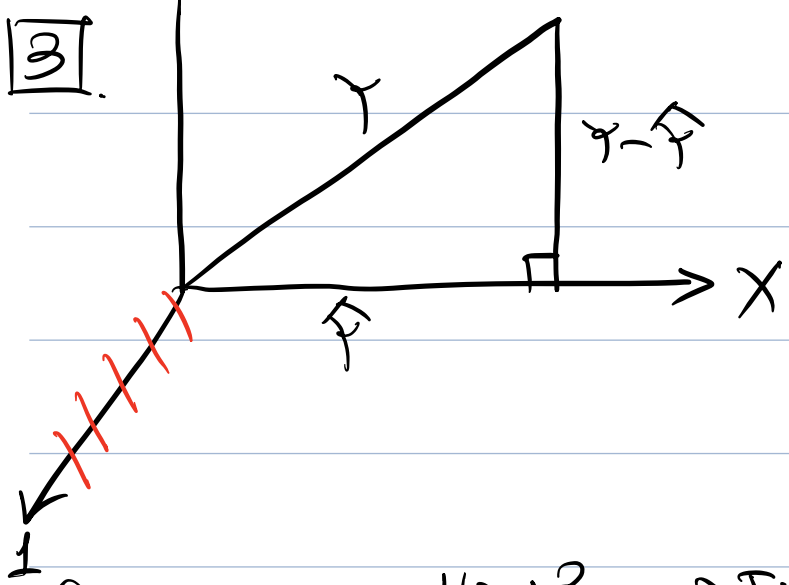
$\boxed{3}$.



(a) $SST = \|Y\|^2 = Y^T Y$

$SSE = \|Y - \hat{Y}\|^2 = Y^T (I - H_X) Y$

$SSR = \|\hat{Y}\|^2 = Y^T H_X Y$

$\because$ check : $SST = SSE + SSR$

$$Y^T Y = Y^T (I - H_X) Y + Y^T H_X Y$$

$\boxed{//}$

(b) $H_0 : \beta_1 = \cdots = \beta_p = 0$ vs. H_1 : not H_0

$$F = \frac{SSR / p}{SSE / n-p} = \frac{Y^T H_X Y / p}{Y^T (I - H_X) Y / n-p}$$

① $\dfrac{Y^T H_X Y}{\sigma^2} \sim \chi^2\left(p, \dfrac{\beta^T X^T X \beta}{2\sigma^2}\right)$

$\text{rank}(\underbrace{H_X}_{\text{idemp}}) = tr(H_X) = p$, $\frac{1}{2} \frac{\mu^T H_X \mu}{\sigma} = \dfrac{\beta^T X^T X \beta}{2\sigma^2}$

② $\dfrac{Y^T(I-H_X)Y}{\sigma^2} \sim \chi^2(n-p)$

$$\text{rank}(\underbrace{I-H_X}_{\text{idemp}}) = tr(I-H_X) = n-p$$

$$\frac{1}{2}\frac{\mu}{\sigma}(I-H_X)\frac{\mu}{\sigma} = \frac{1}{2}\frac{\mu}{\sigma}(X-X)\frac{\beta}{\sigma} = 0$$

③ Independence $\quad H_X(I-H_X) = H_X - H_X = 0$

$$F = \frac{\frac{SSR}{\sigma^2}/p}{\frac{SSE}{\sigma^2}/n-p} \begin{cases} \overset{H_1}{\sim} F\left(p, n-p, \frac{\beta^T X^T X \beta}{2\sigma^2}\right) \\ \overset{H_0}{\sim} F(p, n-p) \end{cases}$$

$\blacksquare$

4.

(a) $$\max_a \frac{(a^T\hat{\beta})^2}{s^2 a^T(X^TX)^{-1}a} = \frac{\hat{\beta}^T X^T X \hat{\beta}}{s^2}$$

$$\frac{\partial}{\partial a}\frac{(a^T\hat{\beta})^2}{\underbrace{a^T(X^TX)^{-1}a}_{\text{scalar}}} = \frac{[a^T(X^TX)^{-1}a]\,2(a^T\hat{\beta})\hat{\beta} - 2(a^T\hat{\beta})^2(X^TX)^{-1}a}{[a^T(X^TX)^{-1}a]^2}$$

$$= 0$$

$$\Rightarrow [a^T(X^TX)^{-1}a]\,\hat{\beta} = (a^T\hat{\beta})(X^TX)^{-1}a$$

$$\Rightarrow a = \underbrace{\frac{a^T(X^TX)^{-1}a}{a^T\hat{\beta}}}_{=:c} X^TX\hat{\beta}$$

$$= c X^T X \hat{\beta}$$

$$\max_a \frac{(a^T \hat{\beta})^2}{s^2 a^T (X^T X)^{-1} a} = \frac{c^2 (\hat{\beta}^T X^T X \hat{\beta})^2}{s^2 \cdot c^2 \hat{\beta}^T X^T X \hat{\beta}}$$

$$= \frac{\hat{\beta}^T X^T X \hat{\beta}}{s^2}$$

$\boxtimes$

(b) Under $H_0: \beta = 0$

$$\frac{\hat{\beta}^T X^T X \hat{\beta}}{p s^2} = \frac{Y^T H_X Y / p}{SSE / n-p}$$

$$= \frac{Y^T H_X Y / p}{Y^T (I - H_X) Y / n-p} \quad H_X (I - H_X) = 0$$

$$= \frac{\chi^2(p) / p}{\chi^2(n-p) / n-p}$$

$$\sim F(p, n-p)$$

$\boxtimes$